

PPM Framework

Hand-Solved MILP Test Cases

Modifications: Dynamic Horizon + Large B_1

Internal Technical Reference

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Contents

1 Hand-Solved MILP Test Cases

All cases share these conventions unless explicitly relaxed:

- $\gamma = 1.0$ (no discounting), to keep arithmetic clean.
- $B_1 = 3 \cdot \sum_i \text{BAC}_i$ (large; cash never binding).
- $\bar{\eta}_{i,t} = \bar{\eta}$ constant and deterministic.
- Equal milestone weights: $\phi_{i,j} = 1/M_i$.
- $\alpha_i = 0, \rho_i = 0$ (no advance, no retention) unless stated.
- $\delta^{\text{rec}} = 0.25, \psi = 0.10$ (FIDIC defaults).
- $\pi_i = 0$ unless stated ($CP_i = \text{BAC}_i$: baseline with zero margin).
- Monetary units are abstract (“k”).
- S-curve shape: $a_i = b_i = 2$ (symmetric Beta) throughout.

1.1 Single-Project Cases

1.1.1 SP-1: Perfect Efficiency, No Termination Risk

Economic claim. With $\eta = 1$, no budget pressure, and $\gamma = 1$, the MILP front-loads spend to certify milestones as early as possible. No termination conditions fire.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 4, \quad \text{BAC}_1 = 100, \quad CP_1 = 120, \quad \pi_1 = 0.20.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 1, \quad \mu_1 = 0.30, \quad \tau_1^{\text{tol}} = 2.$$

$$B_1 = 300, \quad H = 4.$$

Milestone payments.

$$R_{1,1}^{\text{net}} = 0.5 \times 120 = 60, \quad R_{1,2}^{\text{net}} = 0.5 \times 120 = 60.$$

Payment identity: $0 + 60 + 60 + 0 = 120 = CP_1$. ✓

Progress mechanics. $P_{1,t} = \eta \cdot x_{1,t}/\text{BAC}_1 = x_{1,t}/100$ per period. To certify MS1, need cumulative $P \geq 0.5$, i.e. cumulative spend ≥ 50 . To certify MS2, need cumulative $P = 1$, i.e. cumulative spend = 100.

Optimal strategy. With $\gamma = 1$ and cash non-binding, the MILP maximises inflow speed. Spend 50 at $t = 1$ (MS1 certified immediately), then 50 at $t = 2$ (MS2 certified).

t	$x_{1,t}$	$P_{1,t}$	Inflow	Net	B_t
1	50	0.50	60 (MS1)	+10	300 → 310
2	50	1.00	60 (MS2)	+10	310 → 320
3	0	1.00	0	0	320
4	0	1.00	0	0	320

Objective.

$$Z_{L1}^* = (60 - 50) + (60 - 50) = \mathbf{20}.$$

Termination check. At $t = 1$: $P_{1,1}^{\text{plan}} = I_{0.25}(2, 2) \approx 0.156$. $\text{SPI} = 0.50/0.156 = 3.2 \gg 1$. $\text{CPI} = 50/50 = 1.0$. $\text{EAC} = 50 + 50/1 = 100 \leq 1.30 \times 100$. Neither condition fires. ✓

Assertions for pytest.

$$\text{A1. } x_{1,1} = 50 \pm 0.5, x_{1,2} = 50 \pm 0.5, x_{1,3} = x_{1,4} = 0.$$

$$\text{A2. } P_{1,1} = 0.50 \pm 0.01, P_{1,2} = 1.00 \pm 0.01.$$

$$\text{A3. MS1 certified at } t = 1; \text{ MS2 certified at } t = 2.$$

$$\text{A4. Status: completed. No } \tau \text{ decrement.}$$

$$\text{A5. } |Z^* - 20| \leq 1.0.$$

1.1.2 SP-2: Advance Payment Harvesting, Forced Termination

Economic claim. A large advance combined with rapidly decaying efficiency makes the project economically dead within two periods. The MILP receives the advance and accepts termination. The settlement exactly cancels the advance, yielding $Z^* = 0$.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 100, \quad \text{CP}_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5).$$

$$\alpha_1 = 0.30 \Rightarrow A_1 = 30, \quad \delta^{\text{rec}} = 0.25, \quad \psi = 0.10, \quad \rho_1 = 0.$$

$$\Omega_1 = 1, \quad \mu_1 = 0.05, \quad \tau_1^{\text{tol}} = 1.$$

Efficiency profile (geometric decay):

$$\bar{\eta}_{1,t} = \begin{cases} 1.00 & t = 1 \\ 0.05 & t \geq 2 \end{cases}$$

Milestone payment computation. Recovery threshold $\psi = 0.10$; MS1 weight $\phi_1 = 0.5 \geq 0.10$, so recovery starts at MS1.

$$R_{1,1}^{\text{net}} = 0.5 \times 100 - 0.25 \times 0.5 \times 100 = 50 - 12.5 = 37.5$$

$$R_{1,2}^{\text{net}} = 0.5 \times 100 - 0.25 \times 0.5 \times 100 = 37.5$$

Total recovered via deductions = 25. Advance outstanding at full completion = 5. (The remaining 5 is recovered at final settlement; identity: $30 + 37.5 + 37.5 - 5 = 100$.)

Feasibility of reaching MS1 from $t = 2$. To reach $\theta = 0.5$ starting from $P = 0$ with $\eta = 0.05$:

$$x^{\text{needed}} = 0.5 \times \text{BAC}/\eta = 0.5 \times 100/0.05 = 1000 \gg \text{BAC}.$$

MS1 is physically unreachable from $t = 2$.

Termination under zero spend.

- $t = 1, x = 0$: $P = 0$. $P_{1,1}^{\text{plan}} = I_{1/6}(2, 2) \approx 0.042 > 0$. $\Delta f = \infty > \Omega = 1$ (Cond. 1). $\text{EAC} = \infty > 1.05 \times 100$ (Cond. 2). Counter: $1 \rightarrow 0$.
- Termination executes at $t = 2$ (the period after counter hits 0).

Settlement at $t_1^{\text{term}} = 2$, $P_{1,2} = 0$. Outstanding advance recovery = 0 (no milestones certified).

$$R_1^{\text{term}} = 0 \times 100 - (30 - 0) = -30.$$

Cash flows.

t	$x_{1,t}$	Inflow	Net	Cumul.
1	0	$A_1 = 30$	+30	+30
2	0	0	0	+30
3	0	$R^{\text{term}} = -30$	-30	0

Objective.

$$Z^* = 30 + 0 - 30 = \mathbf{0}.$$

Note: The advance payment is exactly recovered in the settlement. No allocation is ever worthwhile; any spend at $t = 1$ yields progress too small to certify MS1 (given $\eta = 0.05$ from $t = 2$), and the settlement simply grows by $P \times CP$ which is less than the spend. Hence $Z^* = 0$ regardless of the allocation, and the MILP should allocate zero.

Assertions for pytest.

- A1. $\sum_t x_{1,t} \leq 1.0$ (effectively zero spend).
- A2. Project terminates at $t \leq 3$.
- A3. $R_1^{\text{term}} = -30 \pm 1$.
- A4. No milestone certified.
- A5. $|Z^* - 0| \leq 1.5$.

1.1.3 SP-3: Uniform Allocation, High Margin, Completion Optimal

Economic claim. With large profit margin and constant η , completing the project dominates all alternatives (including termination). With $\gamma = 1$, total net cash flow is independent of timing within the feasible set, so any completing allocation achieves Z^* .

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90.$$

$$M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3), \quad \bar{\eta} = 1.0.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2.$$

Milestone payments.

$$R_{1,j}^{\text{net}} = (1/3) \times 90 = 30 \text{ per milestone.}$$

Total inflow = 90. Total outflow (to complete) = 60.

$$Z^* = 90 - 60 = \mathbf{30}.$$

Uniform allocation realisation. Spend $x_t = 10$ per period: $P_{1,t} = 10t/60 = t/6$.

t	x	$P_{1,t}$	Inflow	Net	Cumul.
1	10	1/6	0	-10	-10
2	10	2/6	30	+20	+10
3	10	3/6	0	-10	0
4	10	4/6	30	+20	+20
5	10	5/6	0	-10	+10
6	10	6/6	30	+20	+30

$Z = 30$. Any completing allocation gives the same Z when $\gamma = 1$.

Termination check. SPI ≈ 1 (actual tracks plan with uniform spend); CPI = 1. No conditions fire. ✓

Assertions for pytest.

A1. Total outflow = 60 ± 0.5 .

A2. All three milestones certified.

A3. Status: `completed`.

A4. $|Z^* - 30| \leq 1.0$.

A5. No termination conditions ever simultaneously active.

1.1.4 SP-4: Front-Loading Beats Uniform Allocation ($\gamma < 1$)

Economic claim. With discounting, earlier inflows are worth more. The MILP front-loads all spending to certify milestones at $t = 1$ and $t = 2$, beating the uniform allocation by a computable margin.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 8, \quad \text{BAC}_1 = 100, \quad \pi_1 = 0.40 \Rightarrow CP_1 = 140.$$

$$M_1 = 2, \quad \theta = (0.5, 1), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0, \quad \gamma = 0.95.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad R_{1,j}^{\text{net}} = 70 \text{ each.}$$

Front-loaded allocation. Spend $x_{1,1} = 50$, $x_{1,2} = 50$, rest zero:

$$\begin{aligned}
 Z^{\text{front}} &= \underbrace{\gamma^0 \times 70}_{\text{MS1 at } t=1} + \underbrace{\gamma^1 \times 70}_{\text{MS2 at } t=2} - \gamma^0 \times 50 - \gamma^1 \times 50 \\
 &= 70 + 0.95 \times 70 - 50 - 0.95 \times 50 \\
 &= 70 + 66.50 - 50 - 47.50 = \mathbf{39.00}.
 \end{aligned}$$

Uniform allocation baseline. Spend $x_t = 100/8 = 12.5$ per period. MS1 certified at $t = 4$, MS2 at $t = 8$.

$$\begin{aligned}
 Z^{\text{uniform}} &= \gamma^3 \times 70 + \gamma^7 \times 70 - 12.5 \sum_{t=0}^7 \gamma^t \\
 &= 0.8574 \times 70 + 0.6983 \times 70 - 12.5 \times \frac{1 - 0.95^8}{0.05} \\
 &= 60.02 + 48.88 - 12.5 \times 6.423 \\
 &= 108.90 - 80.29 = \mathbf{28.61}.
 \end{aligned}$$

Since $39.00 > 28.61$, front-loading is strictly better. The MILP must choose front-loading.

Assertions for pytest.

- A1. $x_{1,1} = 50 \pm 0.5$, $x_{1,2} = 50 \pm 0.5$.
- A2. MS1 certified at $t = 1$; MS2 certified at $t = 2$.
- A3. $Z^* \geq 38.5$.
- A4. $Z^* > Z^{\text{uniform}} = 28.61$.

1.1.5 SP-5: Cure-Period Counter — Exact Boundary

Economic claim. A project with zero η from the start ($t \geq 1$) fires both termination conditions at $t = 1$. With $\tau^{\text{tol}} = 3$, the counter reaches zero at $t = 3$ and termination executes at $t = 4$. The settlement exactly cancels the advance.

Parameters.

$$\begin{aligned}
 n &= 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0. \\
 M_1 &= 1, \quad \theta = (1.0), \quad \phi = (1.0). \\
 \alpha_1 &= 0.20 \Rightarrow A_1 = 20, \quad \delta^{\text{rec}} = 0.25, \quad \psi = 0.10, \quad \rho_1 = 0. \\
 \tau_1^{\text{tol}} &= 3, \quad \Omega_1 = 1, \quad \mu_1 = 0.10. \\
 \bar{\eta}_{1,t} &= 0 \text{ for all } t.
 \end{aligned}$$

Termination timeline. Since $\eta = 0$, no progress is possible regardless of spend. $P = 0$ always. $P^{\text{plan}} > 0$ from $t = 1$: Condition 1 fires ($\Delta f = \infty$). $\text{ACWP} = 0$, $t \geq s_1$: Condition 2 fires ($\text{EAC} = \infty$).

t	Both conds. active?	τ^{rem}	Action
1	Yes	$3 - 1 = 2$	Decrement
2	Yes	$2 - 1 = 1$	Decrement
3	Yes	$1 - 1 = 0$	Decrement; counter hits 0
4	—	—	Termination executes

Settlement at $t_1^{\text{term}} = 4$. $P_{1,4} = 0$. No milestones certified; no recovery deductions.

$$R_1^{\text{term}} = 0 \times 100 - (20 - 0) = -20.$$

Cash flows.

t	x	Inflow	Net	B_t
1	0	$A_1 = 20$	+20	$300 \rightarrow 320$
2	0	0	0	320
3	0	0	0	320
5	0	-20	-20	$320 \rightarrow 300$

Objective. $Z^* = 20 - 20 = 0$.

Assertions for pytest.

A1. $\tau_{1,1}^{\text{rem}} = 2, \tau_{1,2}^{\text{rem}} = 1, \tau_{1,3}^{\text{rem}} = 0$.

A2. Termination period = 4.

A3. $R_1^{\text{term}} = -20 \pm 0.5$.

A4. Status: `terminated`.

A5. $|Z^* - 0| \leq 1.0$.

1.2 Multi-Project Portfolio Cases

1.2.1 MP-1: Two Non-Overlapping Projects, Additive Objective

Economic claim. Two projects with no temporal overlap behave independently. $Z^* = Z_A + Z_B$.

Parameters.

Proj.	s	f	BAC	π	M	θ	$\bar{\eta}$
A (SP-1)	1	4	100	0.20	2	(0.5, 1)	1.0
B (SP-3)	5	10	60	0.50	3	(1/3, 2/3, 1)	1.0

$H = \max(4, 10) - \min(1, 5) + 1 = 10$. $B_1 = 3 \times (100 + 60) = 480$.

Since $f_A = 4 < s_B = 5$: zero overlap. Each solves independently.

$$Z^* = Z_A + Z_B = 20 + 30 = \mathbf{50}.$$

Assertions for pytest.

A1. $|Z^* - 50| \leq 1.0$.

A2. Project A: `completed`.

A3. Project B: `completed`.

A4. No terminations.

A5. $x_{A,t} = 0$ for $t \geq 5$; $x_{B,t} = 0$ for $t \leq 4$.

1.2.2 MP-2: Good Project vs. Bad Project (Unconstrained Cash)

Economic claim. With abundant cash, the MILP allocates zero to the bad project (which terminates giving net zero) and maximises spend on the good project. $Z^* = Z_{\text{Good}}$.

Parameters. *Project Bad:* $BAC_B = 100$, $CP_B = 100$, $\pi_B = 0$, $s_B = f_B^{\text{plan}} = 1 \dots 8$, $M = 2$, $\theta = (0.5, 1)$, $\alpha_B = 0.30$ ($A_B = 30$), $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$, $\rho_B = 0$, $\Omega_B = 1$, $\mu_B = 0.10$, $\tau_B^{\text{tol}} = 2$.

$$\bar{\eta}_{B,1} = 1.0, \quad \bar{\eta}_{B,t} = 0.05 \text{ for } t \geq 2.$$

Project Good: $BAC_G = 80$, $CP_G = 120$, $\pi_G = 0.50$, $s_G = 1$, $f_G = 8$, $M = 2$, $\theta = (0.5, 1)$, $\alpha_G = 0$, $\rho_G = 0$, $\Omega_G = 2$, $\mu_G = 0.30$, $\tau_G^{\text{tol}} = 3$, $\bar{\eta}_G = 1.0$.

$$H = 8. \quad B_1 = 3 \times 180 = 540.$$

Analysis of Project Bad. From $t = 2$, $\eta_B = 0.05$. To reach MS1: need $50/0.05 = 1000 \gg BAC$. MS1 unreachable. With zero spend: counter decrements from $\tau^{\text{tol}} = 2$, hitting zero at $t = 3$, termination at $t = 4$. Settlement: $R_B^{\text{term}} = 0 - 30 = -30$. Advance = +30. Net contribution of Bad = $30 - 30 = 0$.

Analysis of Project Good. Front-load: $x_{G,1} = 40$, $x_{G,2} = 40$. $P_{G,1} = 0.5$ (MS1, $R_{G,1}^{\text{net}} = 60$), $P_{G,2} = 1.0$ (MS2, $R_{G,2}^{\text{net}} = 60$). $Z_G = (60 - 40) + (60 - 40) = 40$.

$$Z^* = Z_B + Z_G = 0 + 40 = 40.$$

Assertions for pytest.

- A1. $\sum_t x_{B,t} \leq 2.0$.
- A2. Project Bad terminates at $t \leq 4$.
- A3. Project Good: **completed**.
- A4. $|Z^* - 40| \leq 1.5$.

1.2.3 MP-3: Advance Harvest Then Redirect to Good Project

Economic claim. This repeats MP-2 with a tighter initial balance, verifying that the advance from Bad is correctly absorbed into the shared balance before being spent on Good. The net result is identical because advance cancels settlement exactly.

Parameters. Same as MP-2 but $B_1 = 0.5 \times (100 + 80) = 90$.

Period-by-period trace.

t	x_G	x_B	Inflow	Net	B_t
1	40	0	$A_B = 30$ and $R_{G,1} = 60 = 90$	+50	$90 \rightarrow 140$
2	40	0	$R_{G,2} = 60$	+20	$140 \rightarrow 160$
3	0	0	$R_B^{\text{term}} = -30$	-30	$160 \rightarrow 130$
4-8	0	0	0	0	130

$$Z^* = 50 + 20 - 30 = 40. \quad B_t \geq 0 \text{ throughout. } \checkmark$$

Assertions for pytest.

- A1. $B_t \geq 0$ for all t .
- A2. Project Bad: advance received at $t = 1$; terminates by $t = 4$.
- A3. Project Good: **completed** by $t = 2$.
- A4. $|Z^* - 40| \leq 1.5$.

1.2.4 MP-4: Three Projects, Budget-Driven Termination of Low Project

Economic claim. Project Low has $\eta = 0.5$, so achieving MS1 requires spend that immediately triggers EAC overrun. The MILP terminates Low and completes High and Mid, achieving $Z^* = Z_H + Z_M + Z_L = 30 + 12 + 0 = 42$.

Parameters. $H = 6$, $B_1 = 150$, $\gamma = 1$. All: $\alpha = 0$, $\rho = 0$, $\tau^{\text{tol}} = 2$, $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$.

Proj.	BAC	π	CP	M	θ	$\bar{\eta}$	Ω/μ
High	60	0.50	90	2	(0.5, 1)	1.0	2/0.40
Mid	60	0.20	72	2	(0.5, 1)	1.0	2/0.30
Low	60	0.10	66	2	(0.5, 1)	0.5	1/0.10

Why Low terminates. To reach MS1 ($\theta = 0.5$) with $\eta = 0.5$: $x^{\text{needed}} = 0.5 \times 60/0.5 = 60 = \text{BAC}$. After spending 60: $\text{ACWP} = 60$, $\text{EV} = 0.5 \times 60 = 30$, $\text{CPI} = 30/60 = 0.5$. $\text{EAC} = 60 + (60 - 30)/0.5 = 60 + 60 = 120 > (1 + 0.10) \times 60 = 66$. Condition 2 fires immediately. Even spending less does not help: any x yields $\text{CPI} < 1$ from $t = 1$, so $\text{EAC} > 60$ always. Condition 2 always fires. Condition 1 fires because progress is slower than plan with $\eta = 0.5$. **Low terminates after $\tau^{\text{tol}} = 2$ consecutive periods.**

Settlement for Low. $\alpha_L = 0$, so $A_L = 0$ and $R_L^{\text{term}} = P_L \times CP_L \approx 0$ (minimal progress if zero spend). Net contribution = 0.

Optimal allocation (High and Mid). Spend 30 each at $t = 1$ (certify MS1 of H and M simultaneously):

$$P_{H,1} = 30/60 = 0.5 \Rightarrow R_{H,1}^{\text{net}} = 45, \quad P_{M,1} = 30/60 = 0.5 \Rightarrow R_{M,1}^{\text{net}} = 36.$$

$B_2 = 150 - 60 + (45 + 36) = 171$. Spend 30 each at $t = 2$ (certify MS2 of H and M):

$$R_{H,2}^{\text{net}} = 45, \quad R_{M,2}^{\text{net}} = 36.$$

$$B_3 = 171 - 60 + 81 = 192.$$

$$Z^* = (45 - 30) + (36 - 30) + (45 - 30) + (36 - 30) + 0 = 15 + 6 + 15 + 6 = \mathbf{42}.$$

Assertions for pytest.

- A1. High and Mid: completed by $t = 2$.
- A2. Low: terminated by $t \leq 4$; allocation to Low ≤ 2 .
- A3. $|Z^* - 42| \leq 1.5$.
- A4. $B_t \geq 0$ for all t .

2 Summary and Pytest Assertion Reference

Case	Pattern tested	Outcome	Z^*	Key assertions
SP-1	Perfect η , 2 MS, no discount	Completed	20	$x = [50, 50, 0, 0]$; MS1@t=1; MS2@t=2

Case	Pattern tested	Outcome	Z^*	Key assertions
SP-2	Advance harvest, decay, term	Terminated	0	$x \approx 0$; term $\leq t = 3$; settlement = -30
SP-3	High margin, uniform, $\gamma = 1$	Completed	30	Outflow= 60; 3 MS; no Cond. 1/2 fire
SP-4	Front-load beats uniform, $\gamma < 1$	Completed	39	$x = [50, 50, 0, \dots]$; $Z > 28.61$
SP-5	Cure-period boundary, 3 periods	Terminated	0	$\tau = [2, 1, 0]$; term@t=4; settlement= -20
MP-1	Non-overlapping, additive	Both comp.	50	$Z = Z_A + Z_B$; no interference
MP-2	Good vs Bad, large B_1	Bad term.	40	Alloc to Bad ≈ 0 ; Good completes
MP-3	Advance harvest + redirect	Bad term.	40	$B_t \geq 0$; advance absorbed; $Z = 40$
MP-4	3 projects; Low always terminates	Low term.	42	H&M complete@t=2; Low term $\leq t = 4$

Table 1: Test case summary and assertions.

3 Testing Architecture Notes

Framework. Use `pytest`. Install: `pip install pytest`. All numerical comparisons use `pytest.approx(expected, abs=tol)`.

Fixture pattern. Each case is a `pytest` fixture that builds a hard-coded `CONFIG` dict, calls `generate_portfolio`, `build_and_solve`, and `build_records`, then exposes `sol`, `rec`, `portfolio` to the test function.

Plot flag. Add a `confest.py` flag `--plots`; when passed, each test saves the full per-project and portfolio plots to `test_plots/{case_id}/`. This reuses the existing `plot_portfolio` function from `milp_ppm_test.py` directly.

Tolerances.

Quantity	Absolute tolerance ϵ
Objective Z^*	1.5 (solver rounding + MILP gap)
Allocation $x_{i,t}$	0.5
Progress $P_{i,t}$	0.01
Settlement R^{term}	1.0
Counter $\tau_{i,t}^{\text{rem}}$	exact integer (0 tolerance)